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Forecasting Oil Prices Using Time Series Models: A Comparative Study of ARIMA and Prophet

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GitHub Repository Link: https://github.com/sesharaogurijala/Time-Series-Modelling.git

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1. Introduction

Oil prices play a significant role in the global economy, influencing energy markets, transportation costs, and broader economic activity. Due to their volatility and sensitivity to geopolitical and economic factors, accurately forecasting oil prices has long been an important task in financial and economic analysis. Time series forecasting methods provide powerful tools for modelling historical price patterns and generating predictions about future price behaviour.

This study investigates the modelling and forecasting of oil price time series data using both classical statistical techniques and modern forecasting approaches. The primary objective is to analyse historical oil price data and develop predictive models capable of forecasting future price movements. Two modelling approaches are explored in this study: the Autoregressive Integrated Moving Average (ARIMA) model and the Prophet forecasting model. ARIMA is a widely used statistical model that captures temporal dependencies and autocorrelation within time series data, while Prophet is a more recent forecasting tool designed to model trend and uncertainty in time series with minimal parameter tuning.

The analysis begins with an exploratory examination of the dataset to understand its underlying structure, trends, and variability. Stationarity tests and appropriate preprocessing techniques are then applied to prepare the data for time series modelling. An ARIMA model is subsequently developed using a grid search approach to determine optimal model parameters based on the Akaike Information Criterion (AIC). Forecast accuracy is evaluated using the Root Mean Squared Error (RMSE). To provide a comparative perspective, the Prophet model is also implemented and evaluated using the same dataset.

The results of both models are compared to assess their forecasting performance and suitability for modelling oil price data. The findings provide insights into the effectiveness of traditional statistical methods versus modern forecasting approaches in capturing patterns within financial time series data.

Dataset Description

2.1 Data Overview

The data used for this study is a daily observation of oil prices from September 2024 to February 2026. There are 500 observations, and each observation is recorded as price of oil in USD for a particular date. The main variable of interest is the oil price, which refers to market value of the measured oil per day.

The ordering of the data is critical to time series forecasting models. Thus, the dataset was then sorted in chronological order by date and date variable had been converted into datetime format to

enable time based indexing. Such preprocessing is crucial for time series models, allowing them to correctly understand the temporal dependencies between observations.

The price values of oil in the dataset are highly variable and dependent on market conditions, which is typical for commodity markets. There are numerous factors that impact oil prices such as geopolitical events, supply and demand dynamics, and global economic conditions. This means that studying past price action can tell you a lot about what price may do next.

2.2 Summary Statistics

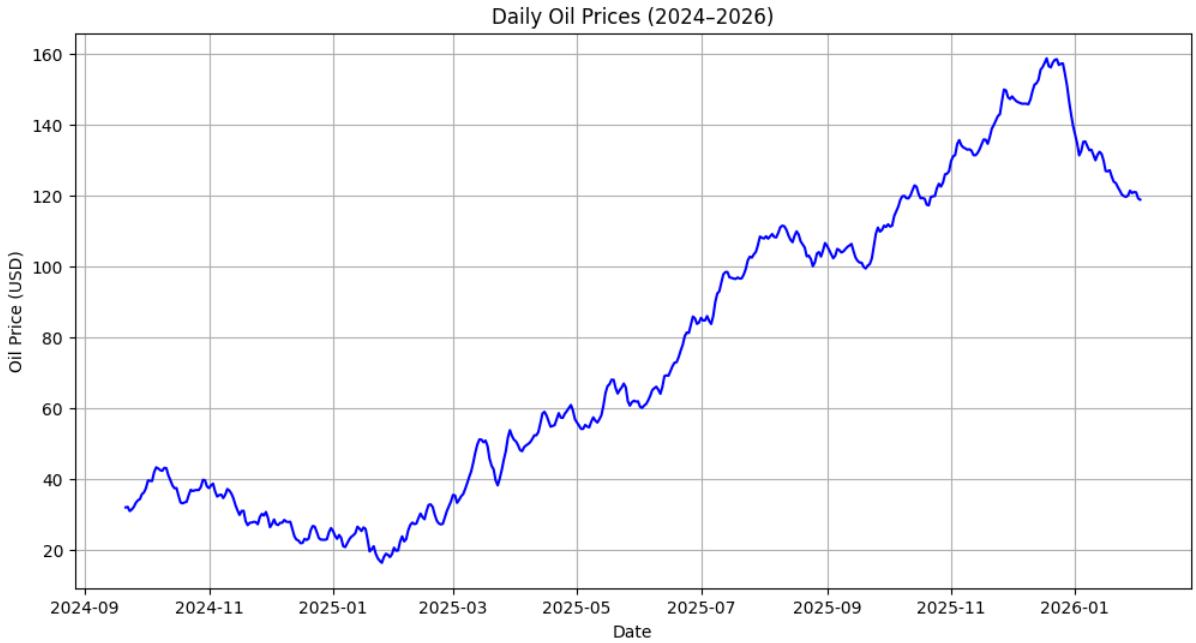
Summary statistics were computed for oil price variable to have first cut insights of the dataset. These metrics summarize the typical and spread of the data.

The dataset has an average of the oil price around $75.87 with a median value of $64.29. The minimum observed value is $16.48 and the maximum is $158.78, which signals a lot of spread in this series of observations. With a standard deviation of around $42.91, this confirms again the huge variability present in the dataset.

As the mean price is greater than median we can say that somewhat of a right-skew exists in our data, as while the prices are less but they occupy large values and stretch upper side of the dataset.

2.3 Time Series Visualization

Figure 1: daily oil price values over the analysed period



A time series plot was created to better understand how the oil price data behaved over time.

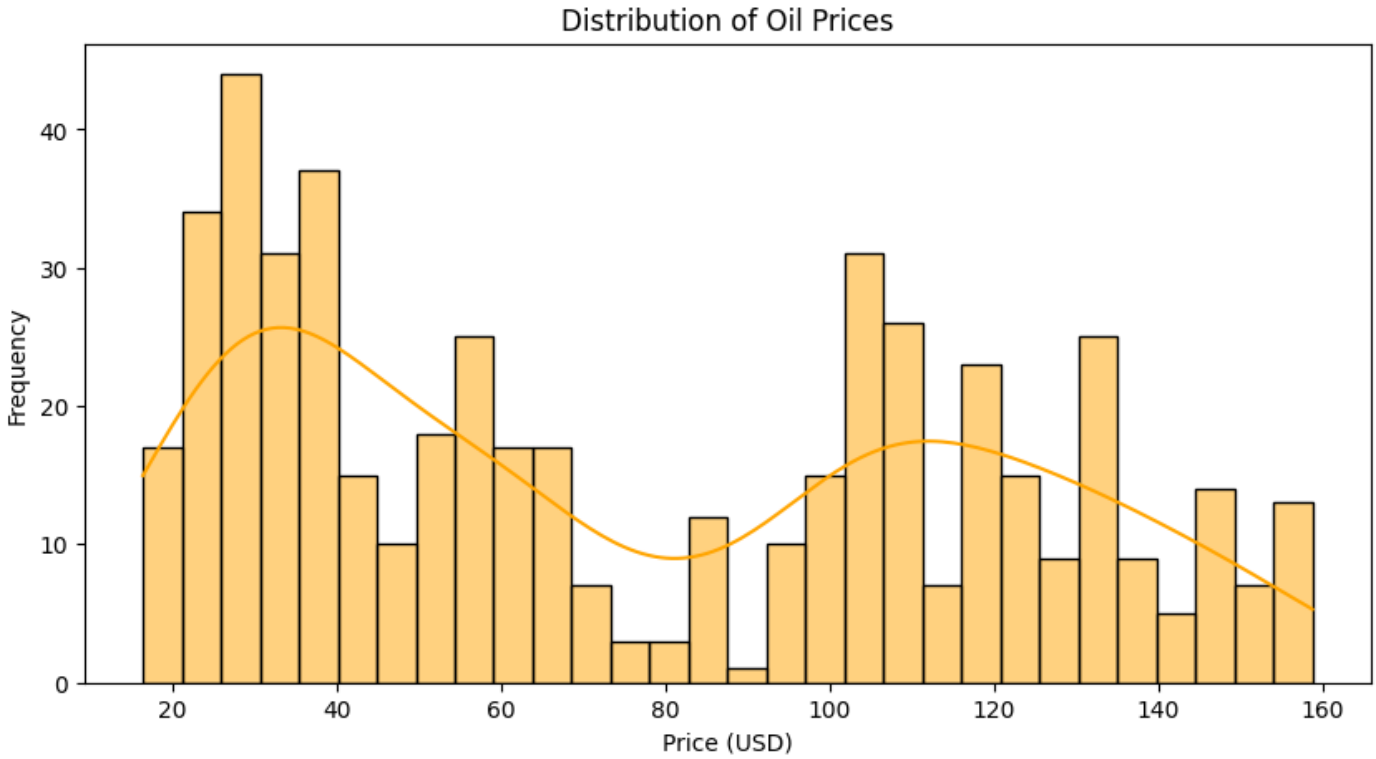
Figure 1 gives the daily oil price values over the analysed period. It shows a distinct positive slope, which means that prices are growing in time. Moreover, the series indicates oscillations around this trend suggesting short-run volatility in the oil market.

predictions on the series, a slight uptrend can be discerned from the time series plot, indicating non-stationarity. Stationarity is the most fundamental assumption of time-series models, which indicates that the statistical properties over time remain unchanged; hence they need to ensure non-stationarity in financial time series [7].

2.4 Distribution of Oil Prices

A histogram of the distribution of oil prices is presented below in Figure 2. It shows the frequency of various price ranges in the data set.

Figure 2: frequency of various price ranges in Dataset



It is evident from the histogram that the distribution is right-skewed with a higher density of observations at lower price levels and fewer observations at higher price levels. The pattern is in line with the summary statistics, indicating some occasional high values within the series.

It helps to assess the distribution of data, find potential outlier(s), and overall variation in series, before exploring advanced time series modelling techniques.

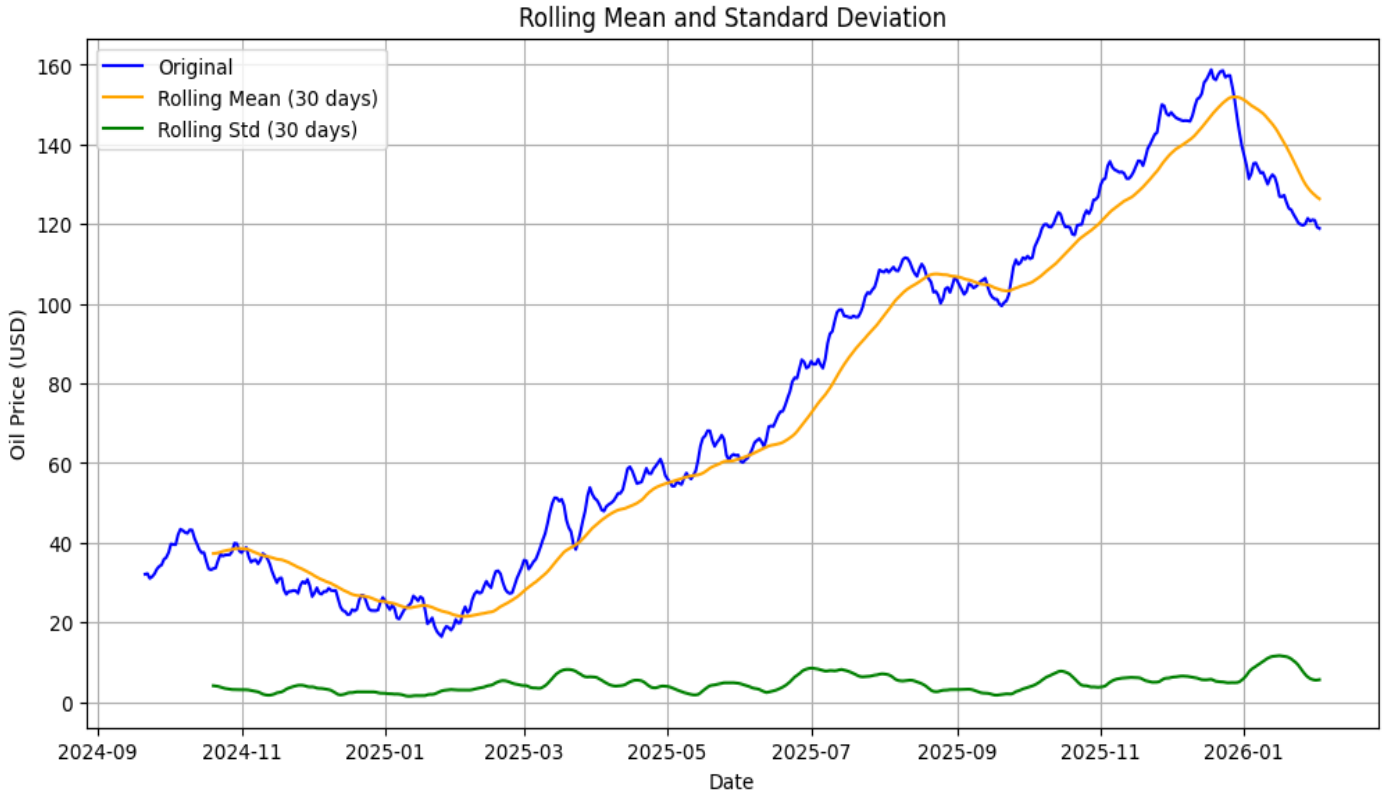
Exploratory Data Analysis

Data from the time series should be cleaned, formatted, plotted, de-trended and smoothed in this step to generate an exploratory data analysis (EDA). EDA gives us insight into the behaviour of oil price series using visual inspection and descriptive analysis as well as potential issues including non-stationarity or volatility.

3.1 Rolling Statistics

Rolling statistics were calculated over a moving window of 30 days to further evaluate the time series stability. The rolling mean and the rolling standard deviation give a dynamic estimate of the average price and how much variation there is over time. If these statistics do not change with time, the series is said to be stationary.

Figure 3: Rolling mean and rolling standard deviation of the oil price series.



We can now plot the rolling statistics as in Figure 3 to show that both the rolling mean and rolling standard deviation change over time. The rolling mean also exhibits an upward trend just like the original series, suggesting a strong trend is present. The rolling standard deviation also, however, fluctuates across the time period of interest; signifying a change in volatility of the data on oil prices.

These observations indicate that the statistical properties of the series are variable in time. Therefore, the oil price series fails to meet the stationarity condition required by some time series models.

3.2 Differenced Series

Since the time series looks non-stationary, one were to apply a common transformation called differencing. Differencing is taking the difference of a time series with its past values in order to remove trends and stabilize the mean of the time series.

Figure 4: Differenced oil price series after applying first-order differencing.

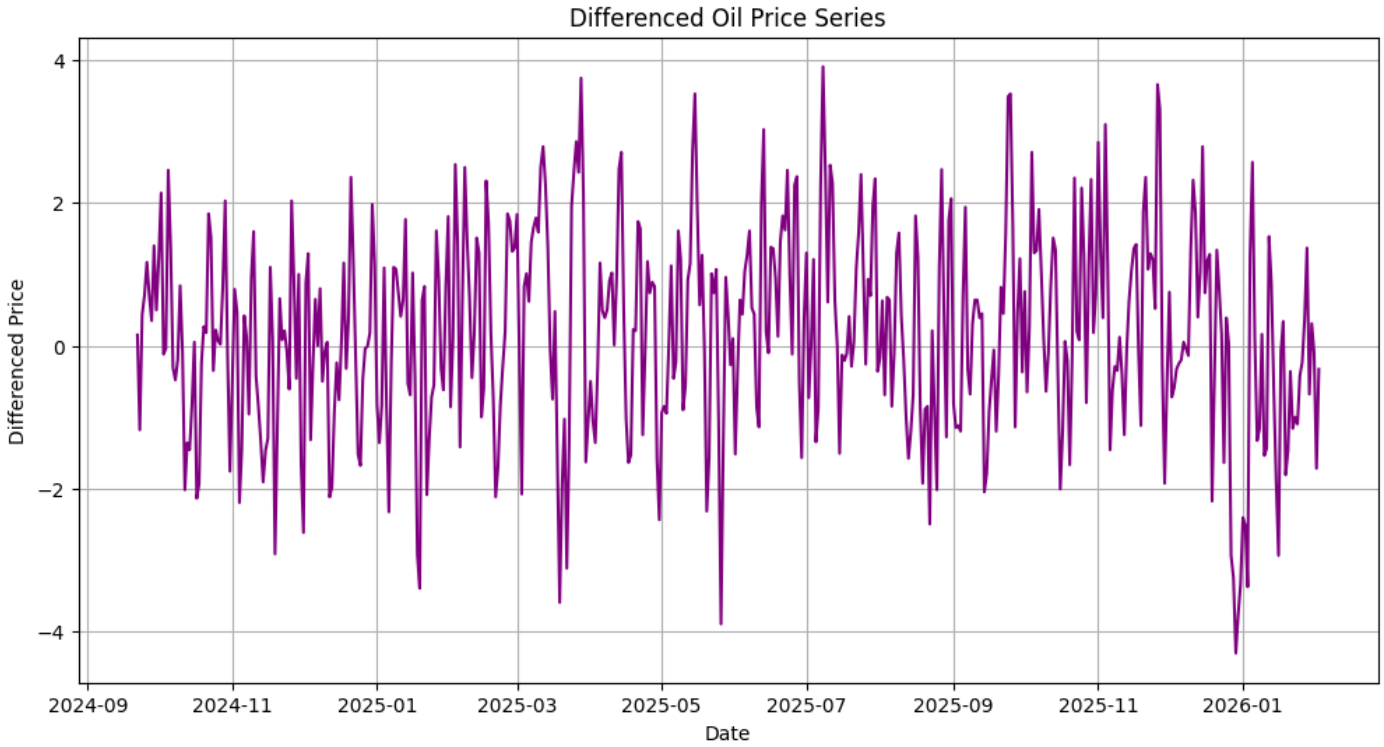


Figure 4 depicts the differenced oil price series. Post differencing, the values oscillates around a static mean (near zero) without an obviously identifiable long-term trend. This means that the differencing transformation has effectively eliminated the non-stationary behaviour i.e.udiant from the original series.

A stationary dataset is a prerequisite to fitting ARIMA models as these models depend on constant Statistical characteristics.

4. Stationarity Testing and Preprocessing

One of the key underlying assumption behind a vast majority of time series forecasting models, is that of stationarity! Stationary Time series has the same statistical properties at any given time such as mean variance and autocorrelation structure. Non-stationarity means that a time series has statistical properties that change over time, which can make forecasting results unreliable.

From the exploratory analysis results shown in previous sections, it can be inferred that oil price series has a clear upward teend and have rolling statistics vary which suggest it may not stationary.

To statistically assess the stationarity of the series, we applied the Augmented Dickey–Fuller (ADF) test.

4.1 Augmented Dickey–Fuller Test

One of the most widely used statistical tests for determining whether a time series is stationary or not is the Augmented Dickey-Fuller test, which operates on this assumption that a time series can have unit root, implying non-stationary (Dickey and Fuller, 1979). Test results produced is shown on the screen above saying null hypothesis of the test that series is non stationary (or unit root exist),alternative hypostasis state that series is staitionary.

Indeed, once the ADF test was applied to the stationary time series of oil price like the original price series VoL08 a p-value higher than 0.05 results thus preventing from rejecting null-hypothesis. It supports the conclusion of original oil price series to be non stationary.

To fix this, we performed differencing on the series. Differencing removes long-term trends in a time series by calculating the difference between each consecutive observation. The transformed series was then subjected to the ADF test again, after differencing.

The differenced series has a p-value much lower than the 0.05 critical level, which leads to rejection of the null hypothesis and concludes that this time series is stationary. This indicates that the transformation has succesfully stabilised price mean and makes the series suitable for ARIMA modelling.

4.2 Autocorrelation Analysis

The next step was to evaluate the autocorrelation structure of the data after transforming the series into a stationary form.] Autocorrelation - The correlation of an observation with previous observations in a time series. This relationship is used to set model hyperparameters in the case of time series forecasting models.

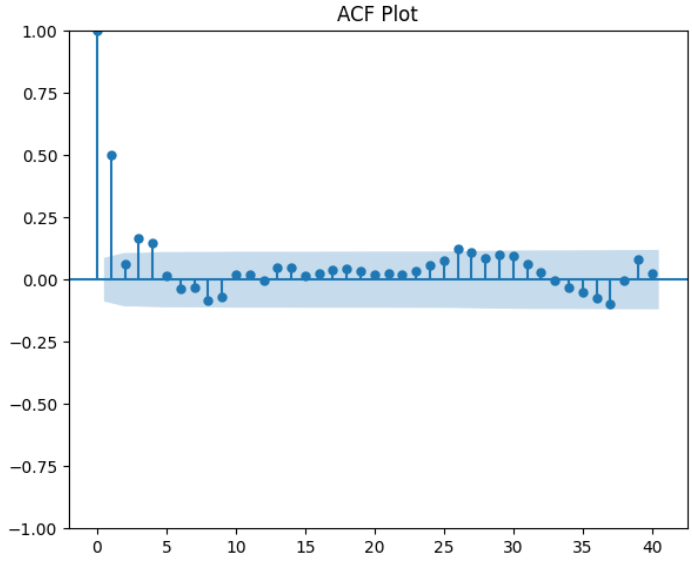
Two important diagnostic tools helped to assess the autocorrelation patterns:

- Autocorrelation Function (ACF)
- Partial Autocorrelation Function (PACF)

These plots are used to determine suitable orders of autoregressive and moving average components for ARIMA models (Box et al., 2015).

4.3 Autocorrelation Function (ACF)

Figure 5: Autocorrelation Function (ACF) plot of the differenced oil price series.



The Auto-Correlation Function plot in Figure 5 shows correlation between the differenced series and its previous values. High peaks in the ACF plot suggest that past values have a significant effect on present data.

The strong autocorrelation in the first few lags that slowly dies down is shown in this plot. This structure indicates the existence of moving average components in the time series model.

4.4 Partial Autocorrelation Function (PACF)

Figure 6: Partial Autocorrelation Function (PACF) plot used to identify autoregressive components.

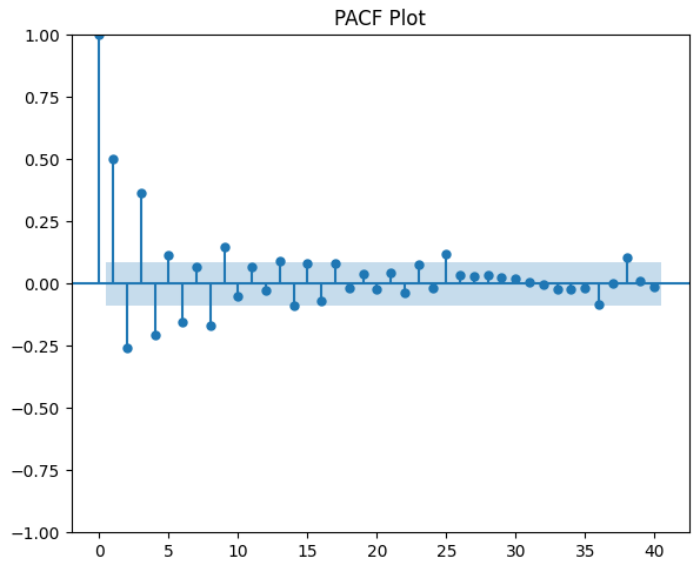


Figure 6: The plot above is the Partial Autocorrelation Function (PACF) for the time series. PACF plot is most commonly used to identify order of autoregressive component in ARIMA models. The plot shows significant spikes for the early lags,type indicating that recent past values are useful in

explaining the behaviour of a time series. This indicates that autoregressive terms might be suitable in the building of the forecasting model.

ARIMA Modelling and Forecasting

5.1 Model Selection

The use of the ARIMA model was appropriate once the oil price series had been converted into a stationary format to enable analysis of time-based dependencies in the data. The ARIMA (AutoRegressive Integrated Moving Average) model is a classical time series model that relies on identifying autoregressive components, performing differencing operations, and including moving average terms based on past observations to enable forecasting from more complex structures of historical patterns in sample data (Box et al., 2015).

An ARIMA model is typically written as ARIMA(p, d, q), where:

- p order of the autoregressive part,
- d is the degree of differencing, d for stationary.
- q is the order of the moving average component.

A grid search approach was implemented to find the best model configuration. We systematically explored the model parameters over ranges:

- $p = 0$ to 8
- $d = 0$ to 2
- $q = 0$ to 8

The Akaike Information Criterion (AIC) was computed for every candidate model. It is widely used in practice to guide model selection by finding a balance between fit and complexity. The best model will have the lowest AIC, reflecting a good trade-off between beneficial variances and unnecessary complexity.

The best-fitting model configuration, determined by the lowest AIC value out of 243 candidate models.

The optimal model identified was:

ARIMA(2,2,7)

The AIC for the best model is about 1429.36.

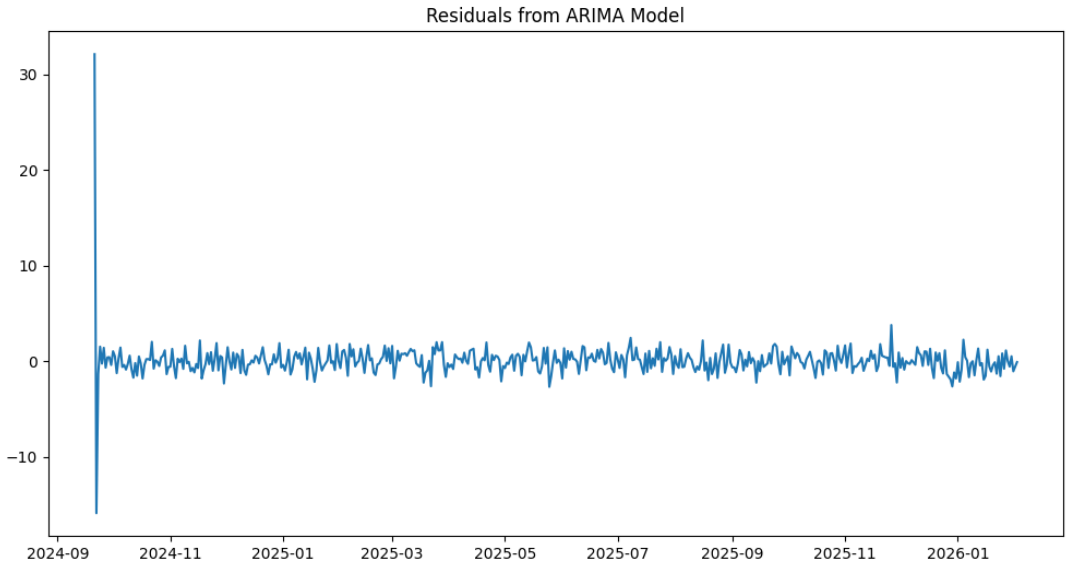
Thus, the selected a model was further fitted and trained for future prediction.

5.2 Residual Diagnostics

Residual diagnostics were then performed on the fitted ARIMA(2,2,7) model, to test whether the model has captured all structure in the time series. After fitting a time series model perfectly, the residuals should behave like white noise; they should be randomly distributed around 0 with no systematic trends.

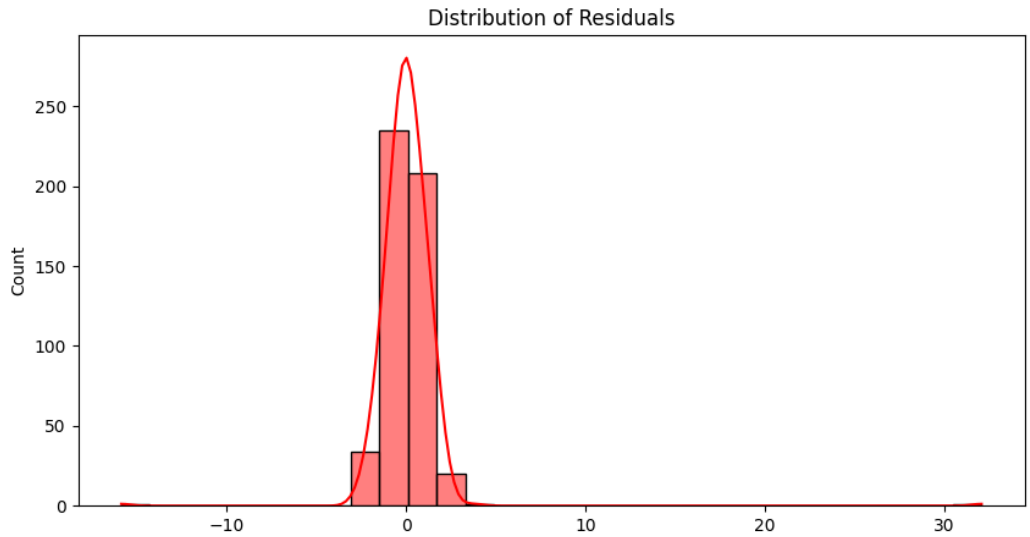
Figure 7 shows the residual time series. The residuals appear to vary randomly about zero with no indication of adjusting trend or systematic structure. This means that the model was indeed able to capture the main trends from our series.

Figure 7: Residual time series obtained from the fitted ARIMA model.



To analyse the distribution of residuals in more detail a histogram was produced (Figure 8). The histogram shows that the residuals are roughly normally distributed and centred around zero. This yields desirable because many statistical models assume normally distributed error.

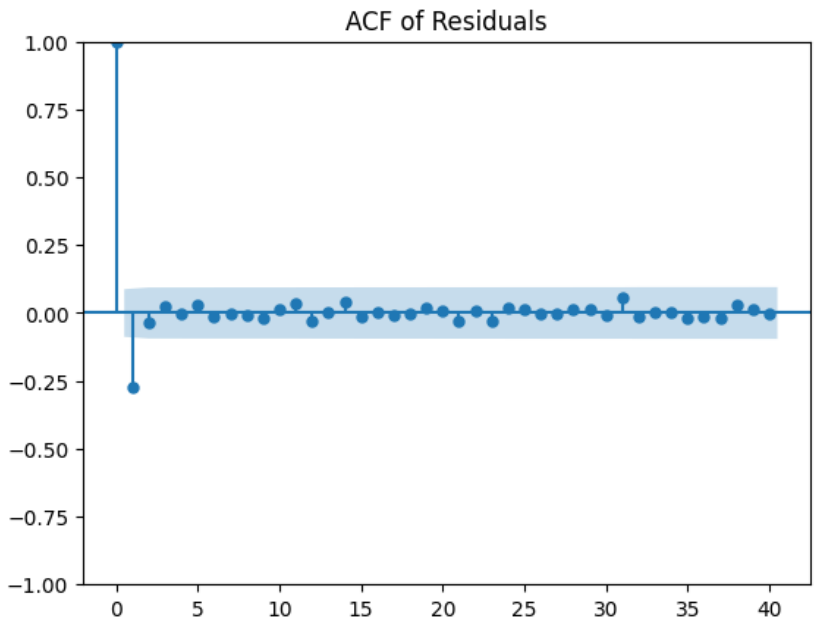
Figure 8: Distribution of residuals from the ARIMA model.



The Autocorrelation Function (ACF) plot was also used to test the residual errors for autocorrelation. Residuals have significant autocorrelation and the temporal structure of the data may not be captured completely by using these regressors.

Figure 9: ACF Plot of Residuals As seen in the plot, most of the autocorrelation values are within significance intervals, represented by the shaded area. This implies that very little significant autocorrelation remains. This suggests that the dependency structure of the time series is sufficiently explained by ARIMA model.

Figure 9: Autocorrelation Function of ARIMA residuals.



5.3 Model Evaluation

The dataset was split into training and testing subsets to assess the predictive ability of ARIMA model. The model was trained on the training data and then used to predict the testing period. The Root Mean Squared Error (RMSE) metric was then used to assess forecast accuracy.

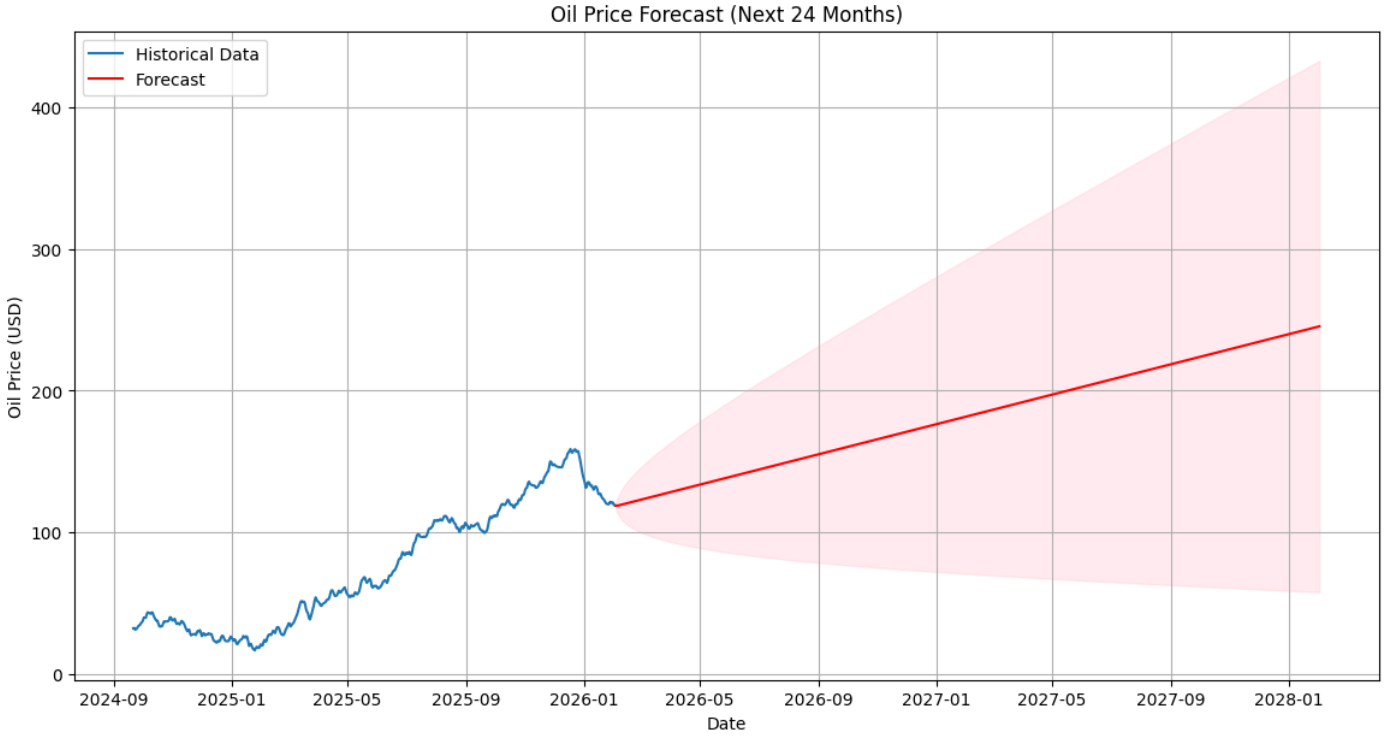
RMSE is used to measure the average magnitude of the errors in a set of predictions, and it is important when evaluating the performance of a forecasting model. The lower the RMSE values that better predictive performance.

The forecasting results of the ARIMA model show an RMSE value of around 14.78, suggesting that on average the forecasted oil prices differ from the actual values by $14.78.

This prediction error is considered reasonable, given the broad scope of oil prices present in dataset and indicates that there are no troubling signs — we have a good fit to true underlying time series behaviour.

5.4 Forecasting Oil Prices

Figure 10: Oil price forecast generated using the ARIMA(2,2,7) model.



Then the fitted ARIMA model was used to forecast oil prices for 24 months ahead. As the dataset is daily in nature when looking to forecast, the horizon was then around 730 days as it would fit the size of training set.

Figure 10 shows the resulting forecast. The plot depicts the historical oil prices, and predicted future values by the ARIMA model. As the forecast horizon tends to reflect predicted oil prices, the upward trend seen in historical data is expected to continue into the previous three months of each year.

The figure also shows confidence intervals, which describe the uncertainty around future predictions it also contains the forecasted values. As we can see, the confidence intervals grow wider for longer forecast horizons, indicating that there is greater uncertainty associated with long time series forecasts.

6. Alternative Forecasting Model: Prophet

6.1 Prophet model overview

A second forecasting method was implemented in addition to the ARIMA model for comparison of modelling performance. This framework, called Prophet (Taylor and Letham, 2018), was chosen for the purpose of forecasting time series. Prophet is an open-source forecasting tool designed for producing high-quality forecasts at scale and has proven particularly useful for generating

predictions of time series that exhibit strong seasonal effects, often seen in business and economics contexts.

Prophet treats a time series as a composition of components like trend, seasonality and irregularities. The core generalization of the Prophet model is to decomposes the time series into additive components describing long-term trends and random variation.

The main feature of Prophet that makes it a great and simple-to-use time series forecasting model is its ability to cope with missing values, trend changes (increasing or decreasing the overall trends) and irregular seasonalities without having to tune too many parameters. This makes it especially useful for real-world forecasting applications where models need to be developed quickly.

Prophet was tested on the same oil price dataset previously used for the ARIMA model, in order to generate a comparative forecast and test if prediction accuracy was improved.

6.2 Prophet Model Implementation

Prophet model requires the data to be formated in a specific way, hence datatable was formatted such that it meets those requirements before fitting the prophet. The Prophet dataset should have two columns as below:

- ds – which is the timestamp (date)
- y – the variable to be forecasted

Modelling concepts would work better on a transactional dataset for the oil price therefore, the oil price time series dataset was converted to this format before training a model. After preparing the dataset, the training of model was done using historical oil prices observations in time series.

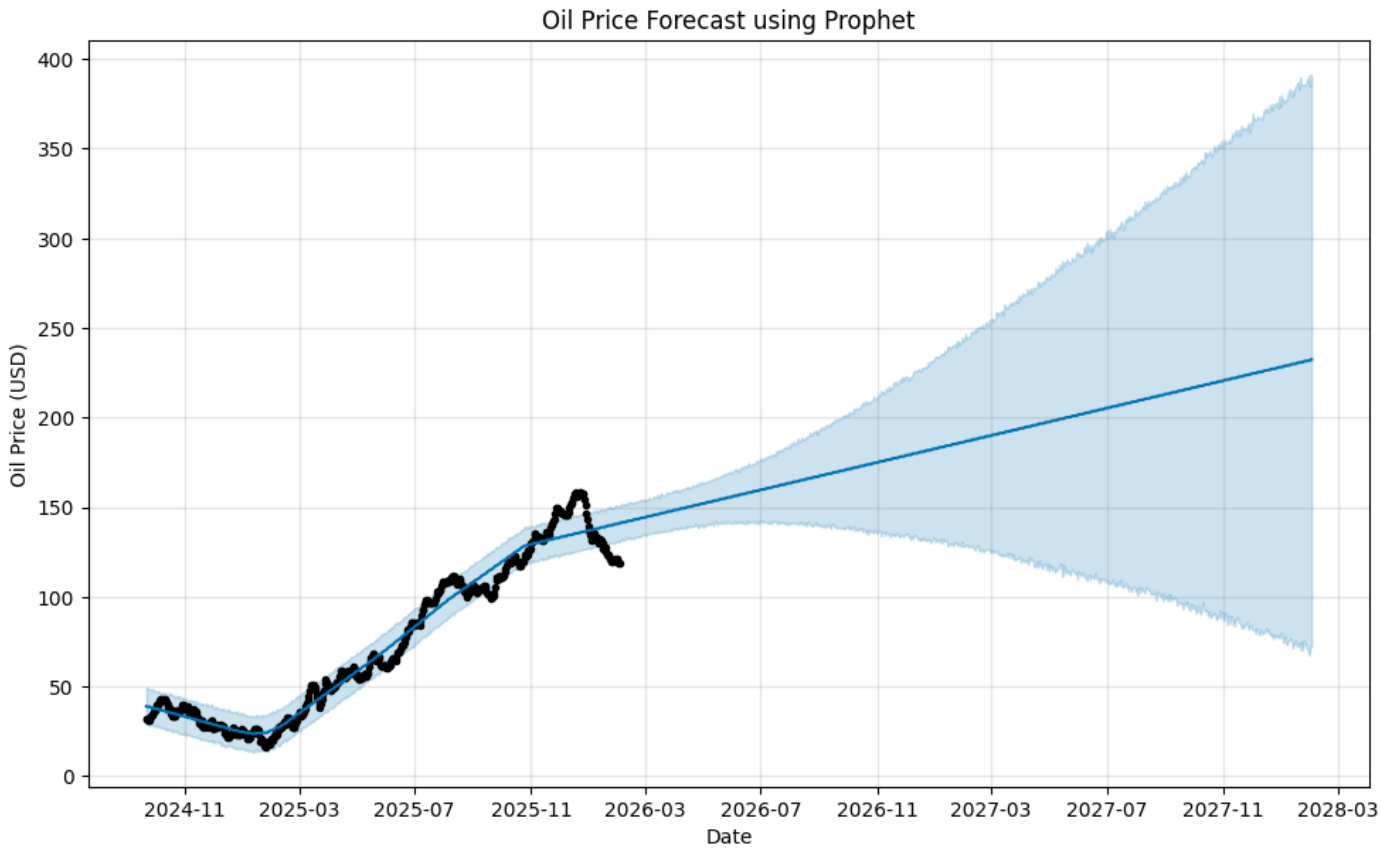
Following model fitting, the future time points corresponding to the forecasting horizon were generated. Prophet model has been used to predict for nearly 730 days equivalent to a forecast horizon of 24 months, like the ARIMA model.

6.3 Prophet Forecast Results

Figure 11 shows the forecast from the Prophet model. Figure showing historical oil prices with predicted values on the future.

The Prophet forecast shows that the price of oil will continue to rise over the forecasting period. Like arima model, the forecasted prices tend to rise. The forecast line is surrounded by a shaded area that indicates the uncertainty interval for the forecasts.

As anticipated, the interval of uncertainty increases with time horizon. This behaviour is a manifestation of the uncertainty that always exists in forecasting time series over long horizons.



6.4 Model Evaluation

The Prophet model was trained and tested on the same train-test set split used to test ARIMA in order to evaluate the predictive performance of this model as well. Again, forecast accuracy was measured by using Root Mean Squared Error (RMSE) metric.

For example, the Prophet model had an RMSE value of around 18.10, which showed a greater prediction error over the ARIMA approach. Indeed, while the Prophet model captured a general upward trend presented in the oil price series of data it still did not predict quite as accurately as ARIMA.

This indicates that although Prophet is able to approximate the general behaviour of the data, there are short time effects in the oil price series, which ARIMA handles better.

Model Comparison

To evaluate the performance of the forecast methodologies deployed in this study, it helped to compare the ARIMA model and Prophet model using quantitative evaluation metrics and qualitative analysis on their forecasting behaviour.

The two models were trained on the same oil price dataset and tested using the same train–test split. The RMSE (Root Mean Squared Error) metric is used to measure the forecast accuracy as a means of quantifying the average magnitude of the prediction errors.

Results of the evaluation are shown in Table 1.

Table 1: Forecast accuracy comparison between ARIMA and Prophet models.

Model	**RMSE**
ARIMA (2,2,7)	14.78
Prophet	18.10

The findings indicate that compared to the Prophet model, the ARIMA model produced a lower RMSE value indicating improved predictive precision. The lower value of RMSE means that the forecasted values using ARIMA model were close to the actual observed oil prices.

The fact that the ARIMA model outperformed the others, may be explained through its way of treating autocorrelation in time series explicitly. ARIMA models reflect relationships between current observations and previous values recorded in the time series, making them useful for temporal structures with short-term dependencies.

On the other hand, Prophet model is designed mainly for modelling long-term trend and overall structure of data. Prophet did well in reflecting the general upward trend of oil prices, but generated less erratic forecasts which failed to align with the short-term variations displayed in the dataset.

Both models yielded similar long-run forecasts, forecasting a continued rise in oil prices over the next 24 months. On a test against historical observations, however, the ARIMA model outperformed the others significantly.

In summary, the results indicate that both of modelling approaches could be used to predict general trend direction of oil price series but the ARIMA was better performing model for this dataset.

Discussion and Future Improvements

This study shows that time series forecasting techniques can work well for modeling and prediction with historical data regarding oil price behavior. The ARIMA and Prophet models were able to incorporate the general upward trends present in the dataset, producing forecasts for the following 24 months. That be said, the ARIMA(2,2,7) model provided a lower RMSE in comparison to the Prophet indicating it is more accurate for this particular dataset. Relative to the oil price series, this indicates that the ARIMA model is better suited to capturing short-term dependencies and linear effects of autocorrelation.

Although the analysis provided valuable insights, there are several limitations to be acknowledged. A potential weakness is the dataset examined in this study has a comparatively short time frame. The dataset includes a total of 500 daily price observations covering around sixteen months of oil

price. Although this was enough to illustrate the modelling techniques, longer historical datasets could capture more complex behaviors such as seasonal variation and long-term economic cycles. A multi-year dataset, therefore, may yield more robust models and forecasts.

One further limitation, which is common here in this study, is the models only depend on historical oil price data. In reality, oil prices are affected by a myriad of external factors including geopolitical developments, global supply and demand conditions, inflation rates, and various macroeconomic indicators. Such explanatory variables could help provide context and potentially enhance predictive performance through inclusion in the forecasting framework.

Further studies might also test more sophisticated prediction methods. For example, Long Short-Term Memory (LSTM) networks and machine learning and deep learning models can potentially model complex nonlinear relationships in time series data and may outperform traditional regression-based models in markets characterized by high volatility. NONETHELESS, hybrid models integrating things like Time Series analysis with machine learning methods are a potential avenue for improving forecasting performance.

In conclusion, although the models implemented in this work present some valuable forecasting tools, future analyses could enhance oil price prediction through the expansion of the dataset, inclusion of external variables and experimentation with more complex modelling approaches.

Conclusion

In this study, we examined the use of time series forecasting techniques in modelling and forecasting oil price movements. The paper used a dataset of 500 daily observations and performed descriptive statistics, stationarity testing, and modelling on the behaviour of the oil price series. The original series was identified as a non-stationary series before implementation of the model because of the apparent strong upward trend and its changing statistical properties requiring that differencing needed to be done.

This study utilized two forecasting methods to be applied: the ARIMA model and the Prophet prediction model. A grid search procedure was conducted to determine the best-predictor ARIMA specification based on Akaike Information Criterion leading us to choose an ARIMA(2,2,7) model. Residual diagnostics showed that the time series structure was well captured by the model. The Mahalanobis algorithm for evaluation of the forecast accuracy was determined that in the case of being used ARIMA model its Root Mean Square Error (RMSE) for matters with whom this method was used amounted to 14.78 what provides practical knowledge on how strong it is predictive performance.

An additional model, Prophet, was also applied as an alternative method for forecasting. Despite successfully capturing the general upward trend of the oil price time series, Prophet did not predict oil prices with as much accuracy: its RMSE was 18.10, higher than that of the ARIMA model.

In short, The ARIMA model yielded better results for this dataset even though both models showed the same trend direction of oil prices. The study showcases the effectiveness of time series forecasting techniques for analyzing and predicting commodity price behavior.

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